

Program 12 – Google Maps Location

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.maplocation">

    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:theme="@style/Theme.MapLocation">

        <meta-data
            android:name="com.google.android.geo.API_KEY"
            android:value="API_KEY" />

        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>

    </application>

</manifest>
```

build.gradle.kts (app)

```
plugins {
    alias(libs.plugins.android.application)
}

android {
    namespace = "com.example.maplocation"
    compileSdk {
        version = release(36)
    }

    defaultConfig {
        applicationId = "com.example.maplocation"
        minSdk = 24
        targetSdk = 36
        versionCode = 1
        versionName = "1.0"
        testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
    }

    buildTypes {
        release {
            isMinifyEnabled = false
            proguardFiles(
                getDefaultProguardFile("proguard-android-optimize.txt"),
                "proguard-rules.pro"
            )
        }
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

dependencies {
    implementation("com.google.android.gms:play-services-maps:18.2.0")
    implementation("com.google.android.gms:play-services-location:21.0.1")
    implementation("com.google.android.material:material:1.11.0")
    implementation("androidx.fragment:fragment-ktx:1.6.2")
}
```

```
}
```

res/layout/activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.fragment.app.FragmentContainerView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/map"
    android:name="com.google.android.gms.maps.SupportMapFragment"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
```

java/.../MainActivity.java

```
package com.example.maplocation;

import android.os.Bundle;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.OnMapReadyCallback;
import com.google.android.gms.maps.SupportMapFragment;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.MarkerOptions;

public class MainActivity extends AppCompatActivity implements OnMapReadyCallback {

    private GoogleMap mMap;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        SupportMapFragment mapFragment = (SupportMapFragment)
            getSupportFragmentManager().findFragmentById(R.id.map);

        if (mapFragment != null) {
            mapFragment.getMapAsync(this);
        }
    }

    @Override
    public void onMapReady(@NonNull GoogleMap googleMap) {
        mMap = googleMap;

        LatLng location = new LatLng(12.9716, 77.5946);
        mMap.addMarker(new MarkerOptions()
            .position(location)
            .title("My Location"));
        mMap.moveCamera(CameraUpdateFactory.newLatLngZoom(location, 15));
    }
}
```